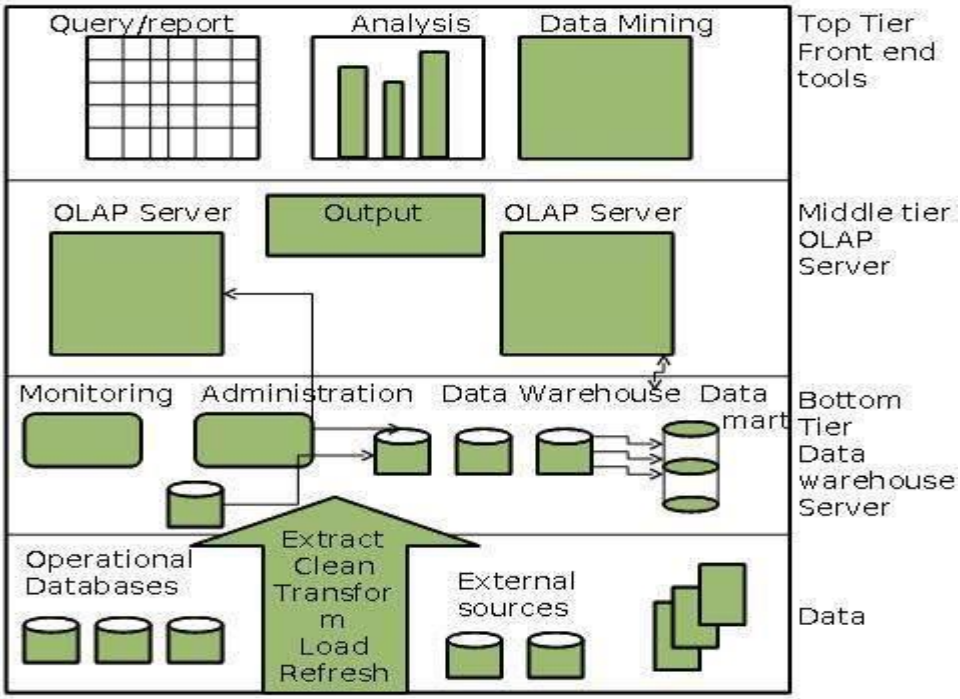
# Unit 3

## Data Warehouse Architecture

Generally a data warehouses adopts three-tier architecture. Following are the three tiers of the data warehouse architecture.

- **Bottom Tier** - The bottom tier of the architecture is the data warehouse database server. It is the relational database system. We use the back end tools and utilities to feed data into the bottom tier. These back end tools and utilities perform the Extract, Clean, Load, and refresh functions.
- **Middle Tier** - In the middle tier, we have the OLAP Server that can be implemented in either of the following ways.
    - By Relational OLAP (ROLAP), which is an extended relational database management system. The ROLAP maps the operations on multidimensional data to standard relational operations.
    - By Multidimensional OLAP (MOLAP) model, which directly implements the multidimensional data and operations.
- **Top-Tier** - This tier is the front-end client layer. This layer holds the query tools and reporting tools, analysis tools and data mining tools.




## Data Warehouse Models

From the perspective of data warehouse architecture, we have the following data warehouse models:

- Virtual Warehouse
- Data mart
- Enterprise Warehouse

**Virtual Warehouse**

The view over an operational data warehouse is known as a virtual warehouse. It is easy to build a virtual warehouse. Building a virtual warehouse requires excess capacity on operational database servers.

**Data Mart**

Data mart contains a subset of organization-wide data. This subset of data is valuable to specific groups of an organization. In other words, we can claim that data marts contain data specific to a particular group. For example, the marketing data mart may contain data related to items, customers, and sales. Data marts are confined to subjects.

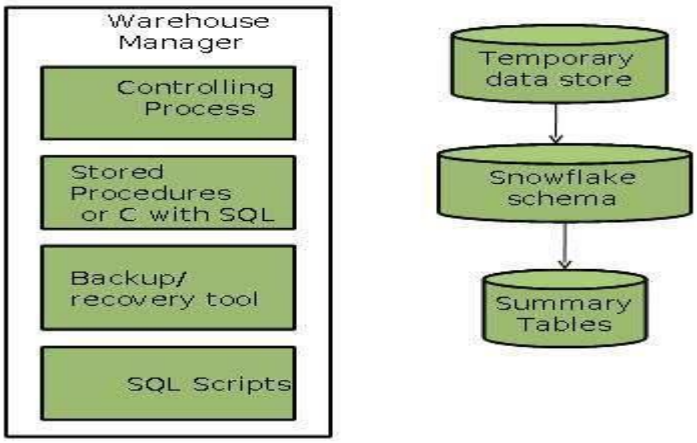
**Enterprise Warehouse**

An enterprise warehouse collects all the information and the subjects spanning an entire organization. It provides us enterprise-wide data integration. The data is integrated from operational systems and external information providers. This information can vary from a few gigabytes to hundreds of gigabytes, terabytes or beyond.

## Warehouse Manager

A warehouse manager is responsible for the warehouse management process. It consists of third-party system software, C programs, and shell scripts. The size and complexity of warehouse managers varies between specific solutions. A warehouse manager includes the following:

- The controlling process
- Stored procedures or C with SQL
- Backup/Recovery tool
- SQL Scripts



**Operations Performed by Warehouse Manager**

- A warehouse manager analyzes the data to perform consistency and referential integrity checks.
- Creates indexes, business views, partition views against the base data.
- Generates new aggregations and updates existing aggregations. Generates normalizations.
- Transforms and merges the source data into the published data warehouse.
- Backup the data in the data warehouse.

# Online Analytical Processing (OLAP)

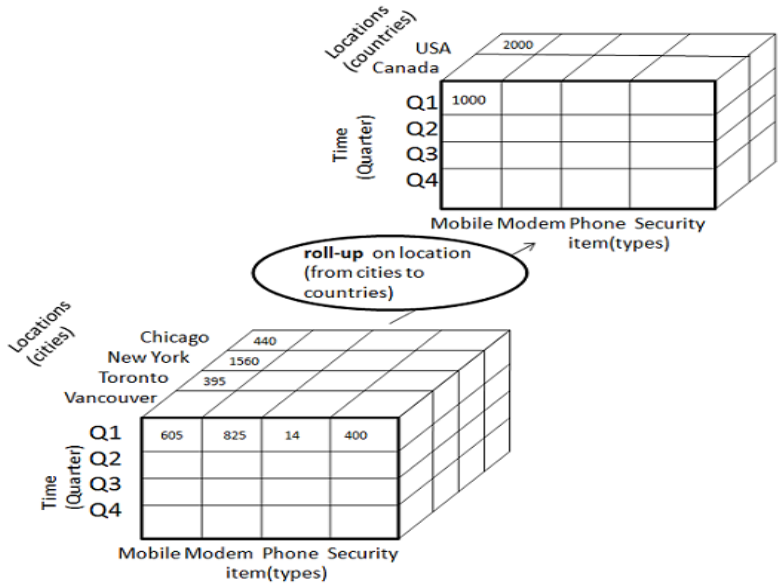
Online Analytical Processing Server (OLAP) is based on the multidimensional data model. It allows managers, and analysts to get an insight of the information through fast, consistent, and interactive access to information.

## OLAP Operations

Since OLAP servers are based on multidimensional view of data, we will discuss OLAP operations in multidimensional data. Here is the list of OLAP operations:

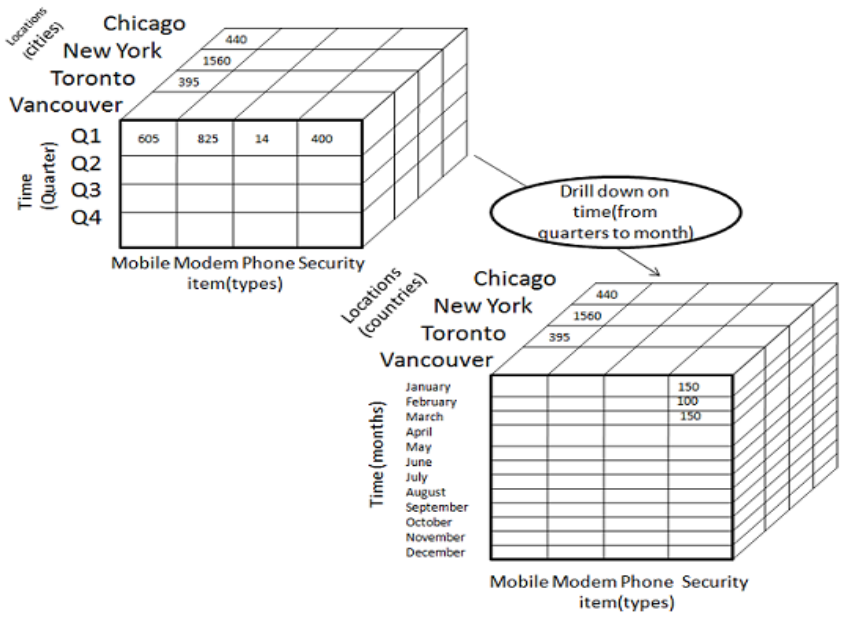
→ Roll-up
→ Drill-down
→ Slice and dice
→ Pivot (rotate)

**Roll-up:** Roll-up performs aggregation on a data cube in any of the following ways: By climbing up a concept hierarchy for a dimension or by dimension reduction. The following diagram illustrates how roll-up works.



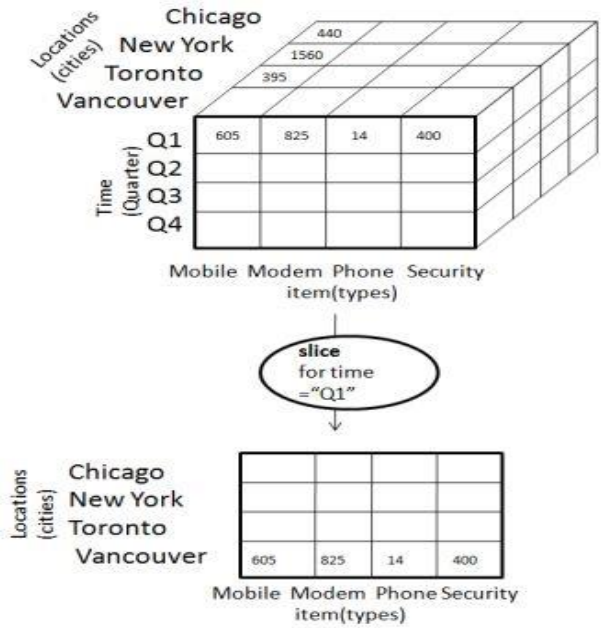
Roll-up is performed by climbing up a concept hierarchy for the dimension location. Initially the concept hierarchy was "street < city < province < country". On rolling up, the data is aggregated by ascending the location hierarchy from the level of city to the level of country. The data is grouped into cities rather than countries. When roll-up is performed, one or more dimensions from the data cube are removed.

**Drill-down:** Drill-down is the reverse operation of roll-up. It is performed by either of the following ways: By stepping down a concept hierarchy for a dimension or by introducing a new dimension. The following diagram illustrates how drill-down works:



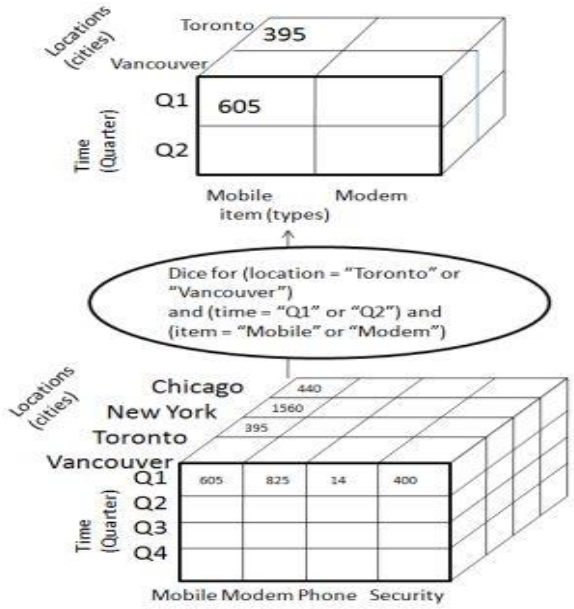

Drill-down is performed by stepping down a concept hierarchy for the dimension time. Initially the concept hierarchy was "day < month < quarter < year." On drilling down, the time dimension is descended from the level of quarter to the level of month. When drill-down is performed, one or more dimensions from the data cube are added. It navigates the data from less detailed data to highly detailed data.

**Slice:** The slice operation selects one particular dimension from a given cube and provides a new sub-cube. Consider the following diagram that shows how slice works.




Here Slice is performed for the dimension "time" using the criterion time = "Q1". It will form a new sub-cube by selecting one or more dimensions.

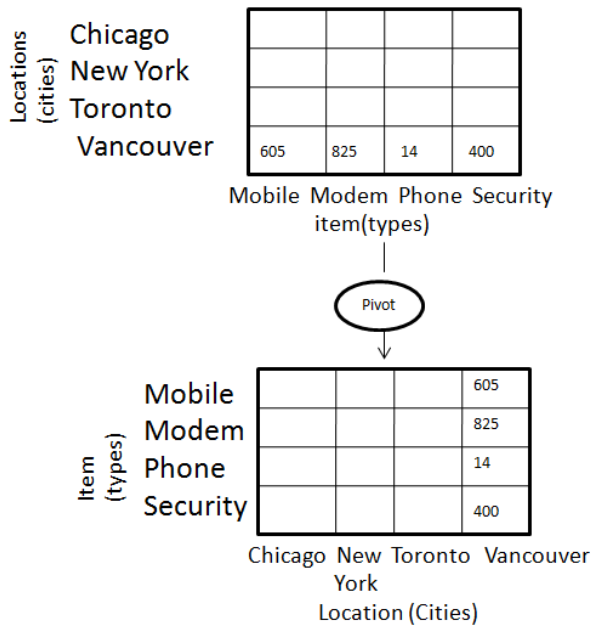
**Dice:** Dice selects two or more dimensions from a given cube and provides a new sub-cube. Consider the following diagram that shows the dice operation.



The dice operation on the cube based on the following selection criteria involves three dimensions.

- (location = "Toronto" or "Vancouver")
- (time = "Q1" or "Q2")
- (item =" Mobile" or "Modem")

**Pivot:** The pivot operation is also known as rotation. It rotates the data axes in view in order to provide an alternative presentation of data. Consider the following diagram that shows the pivot operation.




## OLAP vs OLTP

| OLAP | OLTP |
|---|---|
| Involves historical processing of information. | Involves day-to-day processing. |
| OLAP systems are used by knowledge workers such as executives, managers and analysts. | OLTP systems are used by clerks, DBAs, or database professionals. |
| Useful in analyzing the business. | Useful in running the business. |
| Based on Star Schema, Snowflake, Schema and Fact Constellation Schema. | Based on Entity Relationship Model. |
| Provides summarized and consolidated data. | Provides primitive and highly detailed data. |
| Highly flexible. | Provides high performance. |

## Types of OLAP Servers

We have four types of OLAP servers:

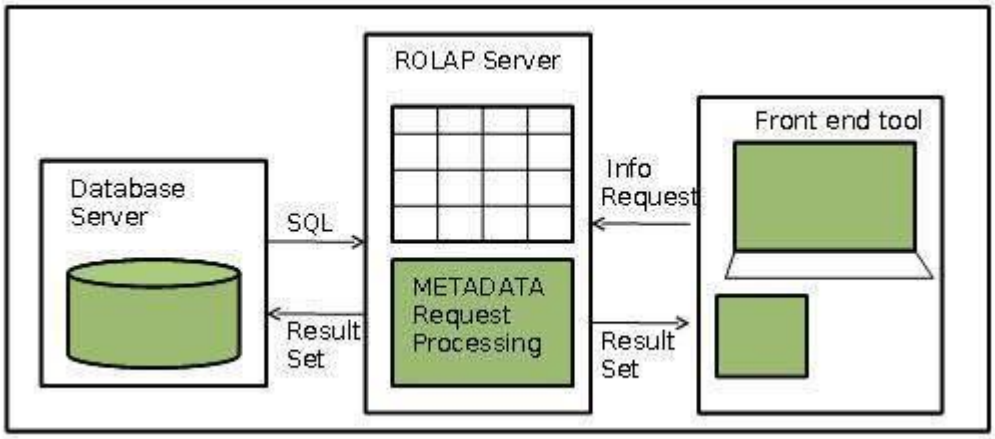
- Relational OLAP (ROLAP)
- Multidimensional OLAP (MOLAP)
- Hybrid OLAP (HOLAP)

**Relational OLAP:-** Relational OLAP servers are placed between relational back-end server and client front-end tools. To store and manage the warehouse data, the relational OLAP uses relational or extended-relational DBMS. ROLAP includes the following:

- Implementation of aggregation navigation logic
- Optimization for each DBMS back-end
- Additional tools and services

ROLAP includes the following components:

- Database server
- ROLAP server
- Front-end tool.




**Advantages**

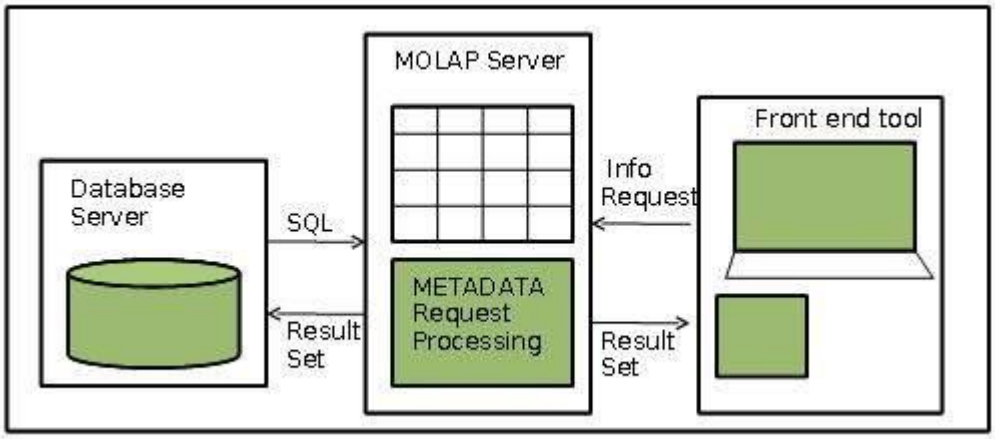
- ROLAP servers can be easily used with existing RDBMS.
- ROLAP tools do not use pre-calculated data cubes.

**Disadvantages**

- Poor query performance.
- Some limitations of scalability depending on the technology architecture that is utilized.

**Multidimensional OLAP:** Multidimensional OLAP (MOLAP) uses array-based multidimensional storage engines for multidimensional views of data. With multidimensional data stores, the storage utilization may be low if the data set is sparse. Therefore, many MOLAP servers use two levels of data storage representation to handle dense and sparse data-sets. MOLAP includes the following components:

- Database server.
- MOLAP server.
- Front-end tool.




**Advantages**

- MOLAP allows fastest indexing to the pre-computed summarized data.
- Easier to use, therefore MOLAP is suitable for inexperienced users.

**Disadvantages**

- MOLAP are not capable of containing detailed data.
- The storage utilization may be low if the data set is sparse.

## Hybrid OLAP (HOLAP)

Hybrid OLAP is a combination of both ROLAP and MOLAP. It offers higher scalability of ROLAP and faster computation of MOLAP. HOLAP servers allows to store the large data volumes of detailed information. The aggregations are stored separately in MOLAP store.

# MOLAP vs. ROLAP

| MOLAP | ROLAP |
| --- | --- |
| Information retrieval is fast. | Information retrieval is comparatively slow. |
| Uses sparse array to store data-sets. | Uses relational table. |
| MOLAP is best suited for inexperienced users, since it is very easy to use. | ROLAP is best suited for experienced users. |
| Maintains a separate database for data cubes. | It may not require space other than available in the Data warehouse. |
| DBMS facility is weak. | DBMS facility is strong. |